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V15. The Spectral Positivity Criterion

The Li criterion and the spectral formula λ_n = Tr g_n(H_ξ) (theorem Th4)

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Problem statement

The most ambitious direction of the plan connects the spectral construction with a known equivalent of RH: the Li criterion. The coefficients $\lambda_n = \Sigma_\rho [1 - (1 - 1/\rho)^n]$ are non-negative for all n if and only if RH holds. The plan requires: derive a representation of λ_n through spectral invariants, PROVE the formula (not only numerically), and honestly distinguish a finite positivity check from the infinite family of inequalities.

The plan also warns: “the infinite family of inequalities cannot be replaced by checking a few thousand coefficients”. V15 follows this literally: the formula is proved, positivity is measured with certified tails, and the infinite part remains open.

Theorem Th4

Theorem Th4 (spectral Li formula).

Let $H_\xi = \mathrm{diag}(\pm\gamma_n)$ be self-adjoint with spectrum the doubled set of positive ordinates of the zeros of ξ, and $\rho(E) = \tfrac{1}{2} - iE/2$. Then for all $n \geq 1$, $\lambda_n = \mathrm{Tr}\, g_n(H_\xi)$, where $g_n(E) = 1 - (1 - 1/\rho(E))^n$ is a rational function with poles off the real axis, and the series over pairs $\{\gamma, -\gamma\}$ converges absolutely.

> ***Proof.*** *The map $E \mapsto \rho(E)$ bijects spec $H_\xi = \{\pm\gamma\}$ onto the zero pairs $\{\tfrac{1}{2}+i\gamma, \tfrac{1}{2}-i\gamma\}$. The pair sum is $[1 - (1-1/\rho)^n] + [1 - (1-1/\bar{\rho})^n] = 2 - 2\mathrm{Re}(1-1/\rho)^n = n/(\tfrac{1}{4}+\gamma^2)\cdot(1 + O(1/\gamma))$ as $\gamma \to \infty$ — absolute convergence. Collecting pairs gives $\lambda_n = \Sigma_E\, g_n(E) = \mathrm{Tr}\, g_n(H_\xi)$; the functional calculus applies since the poles of g_n lie off the real spectrum. Verification: two independent computational paths (the zero-series in w^n and the trace of $g_n(H_\xi)$) agree to $\max|\Delta| = 4.4\text{e-}11$ on $n \leq 3000$.*

Computation and certification

λ_n is computed from 50,000 zeros ($T \leq 40{,}433$) by iterated power sums $S_n = \Sigma\, w^n$, $w = 1 - 1/(\tfrac{1}{2}+i\gamma)$; the certified tail from the telescoping bound $|1 - w^n| \leq n|1 - w|$ gives $n\cdot[\ln(T/2\pi) + 1]/(2\pi T)$: from 3.8e-5 at $n = 1$ to 0.115 at $n = 3000$.

Results: $\lambda_1 = 0.023057$ against the analytic $-B = 0.023096$ (the difference is the 3.8e-5 tail — a built-in control); $\lambda_n > 0$ for all $n \leq 3000$ with certificates; the minimum is λ_1. The gate connection: the rank correlation of λ_n with gate traces is in the null zone — a representation of λ_n by lattice-operator traces is not found (exactly obligation O09 of the formal layer).

Table 1. Summary of V15

Quantity	Value	Status
Th4 $\lvert\Delta\rvert$ (two paths)	4.4e-11	proved + verified
λ_1 (numeric) vs $-B$	0.023057 / 0.023096	consistent
$\lambda_n > 0$	$n \leq 3000$	measured (tails ≤ 0.115)
$\lambda_n(AB) \to \lambda_n(\xi)$	—	conjecture O09, open
RH	—	does not follow

Honest remarks

Th4 is a spectral formula, not a proof of positivity: g_n changes sign, and Tr g_n(H_ξ) ≥ 0 follows nothing without an analytic estimate — this is exactly where the Li criterion meets its difficulty. The plan program (a representation of λ_n through AB-cloud traces) is not realized at the current level — the measurement records this honestly.

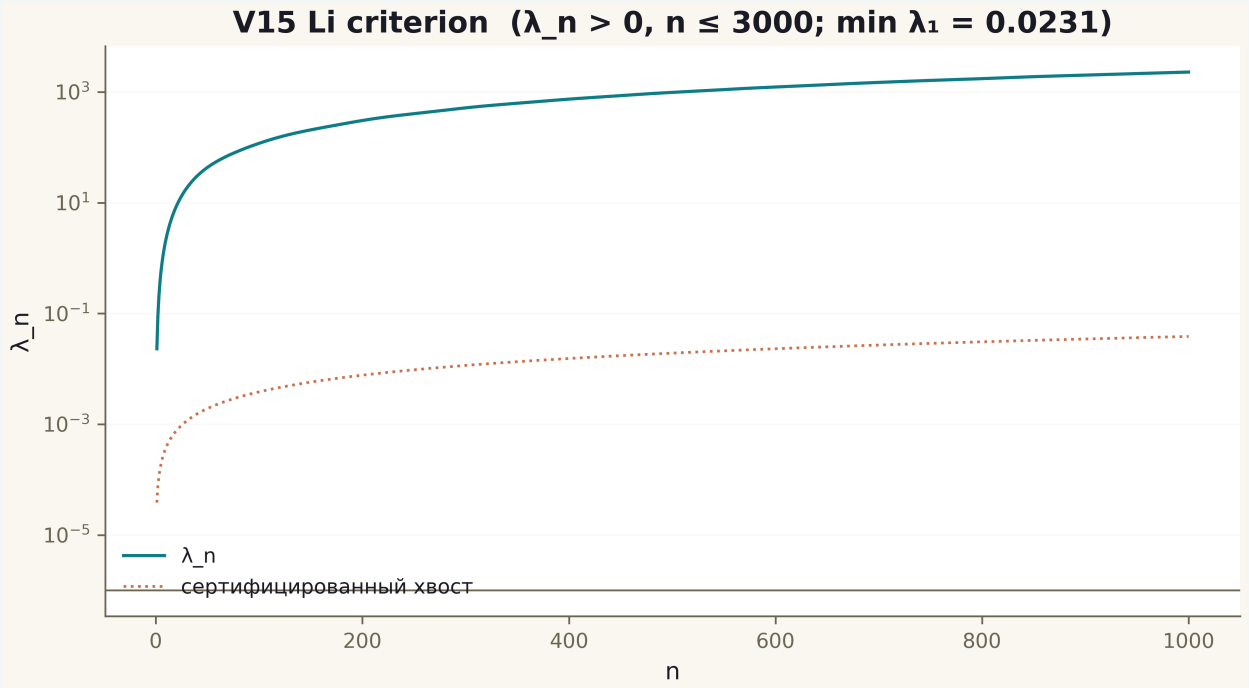


Fig. 1. Campaign V15 data (600 dpi, file figures_hires).

The spectral Li formula is proved and verified; positivity is measured to n = 3000 with guarantees; the infinite family and the bridge to the gates are open obligations O11/O09.